LOSING MY MARBLES!

"Flick Good, Flick for all the marbles.."

<u>KEITH ASHLY DOMINGO</u>
FakeThird

<u>ADRIEL NEYRO CARAIG</u>
iskwipi

Version 3

Abstract

Currently just a game about marbles, deck building, messing around, and having fun.

Contents

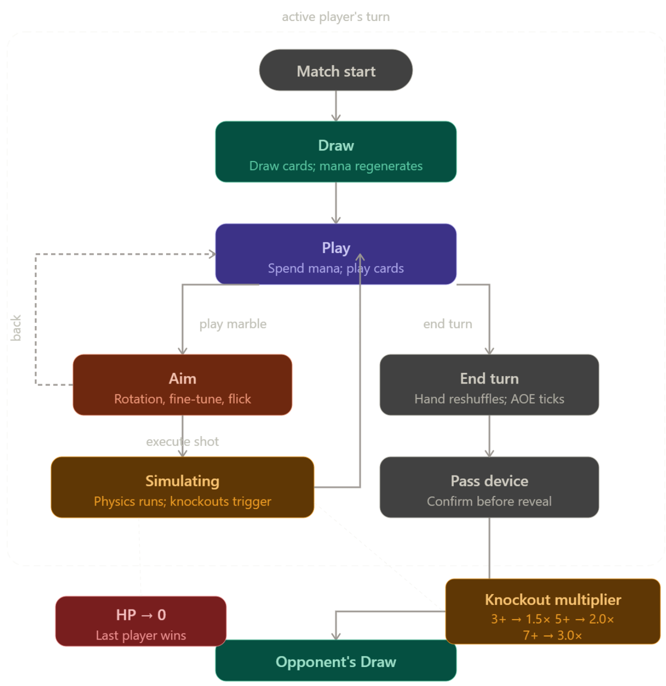
I. Main Concept

Title: **Losing my Marbles!**
Tagline: *"Flick Good, Flick for all the marbles.."*
Hook: A **tactical**, **simulated physics**, **deck builder** reimagining of the Filipino game *'Holen'* that blends precise marble aiming with creative card strategy, where every trick causes mayhem and every shot causes chaos!

II. Core Gameplay Loop

a) **DRAW** → Players draw cards from their private deck; mana regenerates automatically based on the character's mana stat.
b) **PLAY** → The player spends mana to play cards that alter the field or prepare a marble for the upcoming shot. From here, the player chooses one of two branches:
   i) Advance to AIM to take a shot.
   ii) End their turn voluntarily (**END TURN**).
c) **AIM** → The player sets Map Rotation, Fine-Tune Angle, and Flick Slider. A trajectory preview is rendered. The player executes the shot to advance to **SIMULATING**.
d) **SIMULATING** → The player **ends their turn** while the cards they did not use are reshuffled back to their deck and any remaining mana is discarded
e) **END TURN** → Reached only from **PLAY** via Branch B. Unplayed cards reshuffle; mana is discarded; AOE durations tick down. The next player's **DRAW** begins.

Note: Bullets b and c can be done multiple times per turn



Frame A. Mockup In Game GUI by Keith Ashly Domingo

III. Player Goals

A. Before a match:

- Choose a character whose Mana and Power stats fit the intended strategy.
- Build a private deck (any card type) with a clear win condition.
- Build a public marble deck that populates the shared field pool at match start.

B. During a match:

- Knock opposing marbles off the field to trigger their passive effects.
- Protect your own health from reaching zero.
- Manipulate field physics using Terrain and AoE cards to set up chain shots.

Win Condition: Last player with health remaining wins.

IV. Game Elements

A. The Map

- The main playing field where marbles are placed and shot at
- Has variables that affect how shots travel and how marbles interact with each other
- Properties: shape, depth, surface friction, wind resistance

B. The Character

- The unit in which the player plays as
- Has exclusive cards that the player can add to their deck
- Properties: health, mana, power, exclusive cards

C. The Deck

- The set from which cards are drawn during a match
- Is set up by the player before a match using cards from their collection
- Refreshes once all cards have been used
- Split into two: public and private

i. Public Deck

- Is only made up of marble cards
- At the start of a match, the public decks of all players are combined into a single pool of cards from which the marbles are drawn to put into the playing field every time the field is empty

ii. Private Deck

- Can be made up of all card types
- Is the deck from which the player draws cards per turn

D. The Cards

- The main unit used for building a strategy
- Has a variety of effects to either provide advantage or disrupt opponents
- Categorized into four: marble, power-up, trick, or terrain card

E. The Marble Cards

- The main unit used for executing shots
- Interacts with the map and other marbles accordingly
- Behaves differently when in the playing field and when being played for a shot
- Properties: weight, friction, passive effect, active effect

i. When in player's hand

- The active effect of a marble card is applied to the shot the player is currently preparing

ii. When in the playing field

- The passive effect of a marble card is triggered when that marble is knocked off the playing field either by a shot or by other card effects

F. The Non-Marble Cards

- The units that enables the player to build powerful strategies
- Properties: mana requirement

i. Power-up

- Affects the marble that the player currently prepared or the player stats themselves

ii. Trick

- Affects either the marbles currently on the playing field, any of the players' decks, enemy stats and marble, shooting conditions, and everything not in power-up.

iii. Terrain

- Affects the current map condition which affects all players
- Only one terrain card can be active at a time

V. Game Mechanics

A. Match Initialization

- A random map is selected, defining the playing field and establishing the environment conditions using the surface and wind properties of the map
- The public decks of all players are combined into a single pool of cards and the map is set up by drawing a certain number of marbles from the pool and randomly placing them on the playing field
- The turn order for the players will be decided randomly

B. Player Turn

- The player draws a certain number of cards from their deck
- The player generates a certain amount of mana

- The player can play terrain cards which will override the existing terrain, since only one terrain card can be active at once
- The player can play a single marble card that changes the properties of the marble for the current shot being prepared
- The player can play any number power-up and trick cards as long as they have the resources to meet the requirements of those cards
- The player can switch to aiming mode to display and adjust the predicted trajectory of the current shot then finally execute the prepared shot
- The shot is then simulated to produce an outcome of which marbles stayed in the playing field and which marbles were knocked off, then triggering the effects of the latter to wherever they apply
- If there are no more marbles left on the playing field, the game will draw marbles from the pool created in the match initialization to refill the playing field
- If there are no more cards on the player's deck, the game will reshuffle all of the player's consumed cards to be used for that player's next draws
- The player can perform as many shots as they have the resources to do so
- The player can end their turn whenever they desire
- After ending their turn, the player reshuffles their hand back to their deck and discards any remaining mana, then the next player will be able to start their turn

C. Multiplier

Tracks how many marbles exit the field when a marble has been shot:

- 0–2 knockouts → 1.0×
- 3–4 knockouts → 1.5×
- 5–6 knockouts → 2.0×
- 7+ knockouts → 3.0×

Multiplier scales effect values via effect.duplicate() — original .tres resources are never mutated. Resets at the start of each new marble shot.

D. Win Condition

- A player will be eliminated from the match if their health falls down to zero
- A player wins the match by being the last player alive

VI. Game Systems

A. Main UI

- Ensures players have access to their resources while achieving the vibe of the game

B. Physics Engine

- Simulates shot-environment interactions and marble collisions

C. State Machine

- Govern game phases and player turn orders

D. Effect Handler

- Handles marble and card effects for all game elements

E. Inventory Manager

- Manages card collection and deck building

VII. Genre & Inspirations

Primary Genre: **Hybrid Deck Builder**

Categories: Card Games, Simulated Physics, Turn-Based Strategy

Reference Works:

- **Kabuto Park:** The overall vibe and theme and main battle UI
- **Slay the Spire:** The use of selectable heroes with specific cards only available to them, mana usage for card use, and the refresh of the hand per turn unless a card says otherwise.
- **Potionomics:** Cards relating only to certain characters and the deck building aspects, and card design.
- **Peglin:** The mechanic where orb varieties have inherent characteristics and behave differently with their surroundings based established game physics
- **Balatro:** The mechanic where a good setup results into a huge payoff, which is achieved through creative strategies and exciting win affirmations for player satisfaction
- **Pokemon TCGP:** The deck-building system and strategy balancing through proper card variance that produces countless game tactics.

VIII. Feasibility & Risks

A. Feasibility Check

- By Week 5, there should be a working game loop that demonstrates marble shot simulation and deck management, with some game mechanics already implemented and a sample set of playing cards

B. Technical Risks:

i. Physics Implementation

- Simulating environment and marble physics may get out of hand when accounting for too much entities and variables
- Probability of desynchronization with predicted path travel when calculating due to combination of effects and entities affecting the physics of the marble

ii. Game Balancing

- The gameplay and the cards themselves should feel fair but competitive for all players, which might take a lot of time to get right

iii. **Finding Assets**
- As the game is pretty unique in its themes, it will be difficult to find quality assets that achieve a fun yet comforting vibe

iv. **Smart AI Implementation**
- As it will be unfair for the user to be limited to play the game with only having a partner, an implementation of solo play will be required but the challenge will be implementing an AI that the user can compete with.

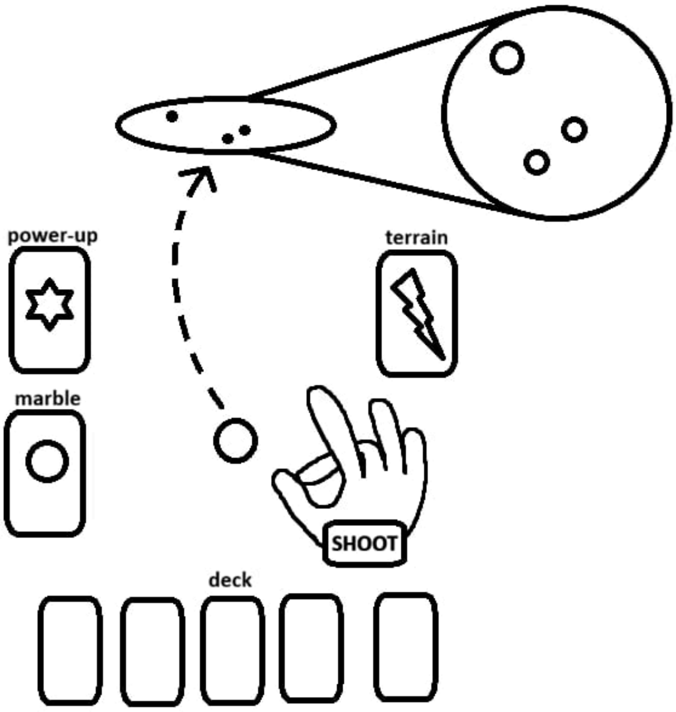
v. **Single Device Turn Based Battle**
- Due to the current scope the game will be limited to just one device where the problem currently resides is in making sure that the turn based aspect still feels fun and not stale.

IX. Scope Management

1. **Minimum Viable Product**
   a. A demo game presenting a complete implementation of all the game mechanics but with 2 sample decks, 2 sample characters, 1 sample map, and PvP gameplay only
2. **Stretch Goals**
   a. Broad selection of cards, characters, and maps
   b. Cohesive UI/UX that achieves the overall theme
   c. Immersive story and balanced game progression
3. **Maximum Final Product**
   a. An online multiplayer mode to battle other players with cards and characters drawn from packs and loot boxes bought using in-game currency earned by progressing through an immersive offline campaign where the player battles the computer with progressive difficulty
4. **Excluded Features**
   a. 3D simulation for marble physics
   b. World exploration or free roam
   c. Real-time action gameplay

X. Mockup Designs

Design A. Mockup In Game GUI by Keith Ashly Domingo



Design B. Mockup Shooting GUI by Adriel Neyro Caraig

XI. Controls Schema

1) **Input** — Action
2) **Drag card toward center** — Play a card from your hand
3) **Flick Slider** — Set how hard you shoot (0 = softest, 10 = hardest)
4) **Map Rotation buttons (hold)** — Spin the entire field to change shot direction

5) **Fine-Tune buttons (hold)** — Make small angle adjustments before shooting
6) **Aim button** — Switch to aiming mode
7) **Execute button** — Fire the marble
8) **Back button** — Return to card-playing without shooting
9) **End Turn button** — Voluntarily end your turn

The game separates big direction changes (Map Rotation) from small precision tweaks (Fine-Tune) to mirror how a real marble player repositions their body then adjusts their wrist. Dragging a card before playing it makes the cost feel tangible. Button-based input was chosen over free mouse-click aiming so the game works equally well on desktop and on a single shared device passed between players.

XII. Prototype Learnings

1) What worked

The physics felt immediately fun. Marbles bounced, rolled, and knocked each other around in satisfying and unpredictable ways from the very first test and no major overhaul was needed. Cards were also easy to create; because each card is just a data file listing its effects, new cards could be added quickly without touching any game logic. The shot preview was clear and updated smoothly. Pass-and-play worked cleanly, correctly hiding each player's hand and requiring a confirmation tap before anything was revealed.

2) What is being cut or reworked

Online multiplayer during active development is being deferred. The online flow gets stuck at the character selection screen and does not proceed to the match. Rather than delay everything else, online play is moved to the final phase — fixing it requires no rewrites since the underlying architecture already supports it.

Per-marble friction differences are also deferred. All marbles currently slide at the same speed because a global field setting overrides their individual values. Restoring marble-to-marble feel differences is a post-submission tuning task.

3) Technical surprises

The card drag-and-play had a hidden timing issue. The card plugin's confirmation signal only fires after a slow animation finishes, which causes the game to miss card plays entirely. This was resolved by hooking into a different point in the plugin. Exit detection also triggered falsely, marbles were being counted as knocked out the moment they spawned. The fix was to only count marbles that had already been confirmed as inside the field first. On the positive side, offline and online code turned out to share the exact same execution path, meaning no separate offline version of the logic was needed and this saved a significant amount of work.

XIII. Updated Scope

The MVP is a complete two-player offline pass-and-play match with all core systems working:

- **the full turn flow of draw play, aim, shoot, and opponent's turn**
- **two playable characters with different stats**
- **over 20 cards across all five card types**
- **a physics field with friction, gravity, and stacking effects**
- **a knockout chain multiplier scaling up to 3×**
- **deck cycling with a discard pile**
- **a one-marble-per-shot rule enforced with a UI hint.**

The online multiplayer skeleton is in place but the full online match flow is deferred to the final phase.

What is not in this build: **online multiplayer match flow, automated testing, and per-marble friction differences.**